

EUIJIN HONG

+1 412-509-2121 | ehong@andrew.cmu.edu | [in](#) LinkedIn | [G](#)itHub

EDUCATION

Carnegie Mellon University

MS, Electrical and Computer Engineering

Pittsburgh, PA

May 2026

- Cumulative GPA: 4.00/4.00
- Relevant Coursework: Introduction to Deep Learning (11785), Advanced Deep Learning (10707), Deep Generative Modeling (18789), Optimization (18660), Deep Learning Systems (10714)

Yonsei University

BS, Electrical and Electronic Engineering

Seoul, South Korea

Feb 2024

- Major GPA: 3.99/4.00 | Cumulative GPA: 3.91/4.00
- Relevant Coursework: Deep Learning Lab, Introduction to Artificial Intelligence, Intelligent Control, Computer Vision, Probability and Random Variables

PUBLICATIONS

DeltaMomentum: A Key-Value based Anisotropic Momentum Update via Delta Rule

ICML 2026 Workshop (Advisor: Dr. Guannan Qu)

- Developed delta rule momentum update by viewing gradients as key-value paired signals, replacing the canonical momentum update.
- Established theoretical analysis on the anisotropic memory transport, fixed point convergence, K-FAC connection, theoretical FLOPs, and μ P scaling.
- Outperformed standard momentum update (SGD/AdamW) in large language modeling performance on 67M/370M scale using Llama2 and FineWeb-edu, achieving 31.35%/19.28% and verifying per-layer mechanistic diagnostics.

Spectral Equalization Minimizes Total Training Energy: A Control-Theoretic Account of Muon's Advantage

ICML 2026 Workshop, Independent Research

- Established framework that views weight eigenvectors as modes, and bridges the connection to H2-optimal control.
- Verified empirically on MNIST dataset on MLP, verifying lower per-layer Gini coefficient on model trained on Muon.

Convolutional Dynamic Rotary Positional Encoding

Interspeech 2026, Independent Research

- Develop a novel positional encoding method that dynamically generate positional offset conditioned on hidden state inputs, for automatic speech recognition (ASR) architectures (e.g. Branchformer).
- Achieve clear improvement (WER) on all LibriSpeech test splits (5.07% $\rightarrow$ 4.95%) and temporal perturbations on speech robust bench (19.6% maximum relative reduction), using 46.8% less learnable positional encoding parameters than RelPos baseline.

RESEARCH EXPERIENCE

FFNs with Full-Dimensional Weight Modulation Improve Transformer

Jan. 2025 - Jan. 2026

InfiniAI Lab, Carnegie Mellon University (Advisor: Dr. Beidi Chen)

- Develop a novel state mixer component integrated to a transformer, introducing context-dependent rank-1 full-dimensional weight modulation.
- Achieve improvement in language modeling perplexity (-0.14) under controlled parameter count using Llama2 backbone with 125M parameters.

Convolutional Dynamic Rotary Positional Encoding

Jan. 2026 - Feb. 2026

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Real-time Neural Network-based Robot Operating System for Teleoperation

Jan. 2023 - Jul. 2023

Computational Intelligence Lab, Yonsei Univ. (Advisor: Dr. Euntai Kim)

- Developed vehicular robot teleoperation system controlled by human pose signals, by lightweighted human pose estimation (HPE) model via weight pruning.
- Lightweighted YOLOv7-based HPE model via branch-wise pruning, in order to reduce end-to-end latency while maintaining the HPE performance.
- Achieved 15.5% less latency by reducing 17.9% of parameter count while obtaining 95.3% teleoperation accuracy.

SELECTED PROJECTS

Full MoE LM Implementation via Custom CUDA/C++ Kernels

Oct. 2025 - Dec. 2025

Deep Learning Systems Course Project, Carnegie Mellon University

- Developed a customized Mixture of Experts (MoE) based Language Model backend using C++ and CUDA, and frontend using Numpy.
- Implemented custom CUDA kernels for top-k operation, router, and MoE modules, as well as full language model including self-attention and tokenizer.

Extending Fourier Number Embeddings to Language Models - 1st Place Project

Jan. 2025 - Apr. 2025

Introduction to Deep Learning Course Project, Carnegie Mellon University

- Developed novel Fourier Number Embedding (FNE) injection mechanism that extract richer numerical representations while maintaining a homogeneous language modeling framework.
- Designed external encoder networks and parallel adapters to inject embeddings into multiple transformer layers.
- Achieved 1st place (highest in-class project score) among 52 project groups, within the cohort of 283 graduate students.

Effective Anomaly Detection Utilizing GAN Discriminator

Jun. 2023 - Jul. 2023

Deep Learning Lab Course Project, Yonsei University

- Developed novel model pipeline for anomaly detection by utilizing dual Generative Adversarial Network (GAN) discriminators as separate training objectives.
- Achieved 99.6% accuracy for classifying in-distribution data and 96.9% for detecting out-of-distribution data in the MNIST dataset.

PROFESSIONAL EXPERIENCE

Teaching Assistant

Pittsburgh, PA

Introduction to Deep Learning (11785) (Instructor: Dr. Bhiksha Raj Ramakrishnan)

Aug. 2025 -

- Operate office hours, recitations, class forum, and hackathons for the course of 400+ students, handling subjects in fundamentals and cutting-edge knowledge on Deep Learning.
- Contribute on course development, especially designing and improving the assignment, which is on speech-to-text sequence translator by Encoder-Decoder Transformer network.

HONORS AND AWARDS

Minister's Award

Dec. 2022

Ministry of SMEs and Startups

- Awarded to the first-place winner in the Small-Scale Entrepreneur Hackathon competition.

Jilli Scholarship

Mar. 2018 - Sep. 2023

Yonsei University

- An academic scholarship given to high-academic performers by Yonsei University.

High Honors

Mar. 2018, Mar. 2020

Yonsei University

- Granted to the top 3% of students within their academic discipline, among a cohort of 319 students.

SKILLS

Areas of Expertise: Representation Learning, Mechanistic Interpretability, Optimizers, Control Theory, Generative Modeling, Reinforcement Learning

Programming Language: Python, CUDA, C/C++, MATLAB, Verilog, JavaScript

Libraries & Frameworks: PyTorch, HuggingFace, NumPy, Pandas